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Optical character recognition based document image quality assessment

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Optical Character Recognition (OCR) systems play a crucial role in digitizing documents. However, their performance significantly deteriorates when handling low-quality images. Even advanced OCR systems struggle if the input is visually or structurally poor. Therefore, achieving high OCR accuracy requires assessing document image quality in terms of how well characters can be recognized, not just visual clarity. In this work, we propose a Document Image Quality Assessment (DIQA) model that predicts OCR accuracy without requiring the actual execution of an OCR engine. To assess document image quality for OCR performance, twelve distinct features are extracted that capture various aspects of sharpness, focus, edge clarity, and structural distortion. Instead of relying on subjective human opinion scores, we generate labels by measuring the actual OCR accuracy using modern engines like PaddleOCR and Keras OCR. These accuracy scores, calculated using the Levenshtein distance, serve as ground truth labels for training. Using the extracted features and corresponding OCR-based labels, we train the machine learning models to learn the relationship between image characteristics and OCR performance. The proposed models are evaluated using statistical metrics such as RMSE, PLCC, and SROCC to determine the most effective predictor. Our experiments demonstrate the importance of using OCR scores as labels, and the results show that our approach yields improved performance compared to existing baseline methodologies.

KEYWORDS

document IQA, goal oriented image quality assessment, Keras OCR, optical character recognition, Paddle OCR, XGBoost regressor

1 Introduction

As the technology continues to evolve, there is an increasing necessity to modernize and digitize the various processes that we encounter in our daily lives. One critical process in this regard is digitization of documents. Digitization has evolved into an ongoing endeavour, extending from individuals and small businesses to large institutions (Clausner et al., 2020). A key enabler of this process is Optical Character Recognition (OCR), a technology that extracts text from document images and converts it into editable digital formats. OCR not only streamlines document editing and organization but also enables efficient storage and quick retrieval of information, playing a pivotal role in modern document management systems. Beyond basic document conversion, OCR technology offers broader societal and operational benefits. For individuals with visual impairments, it enables access to printed materials through text-to-speech applications. In business and administrative settings, OCR automates information extraction, significantly improving the efficiency and accuracy of data entry and management. The digitization of documents fosters environmental sustainability by reducing paper consumption and physical storage space requirements. Additionally, it enhances collaboration by facilitating easy sharing and distribution of digital

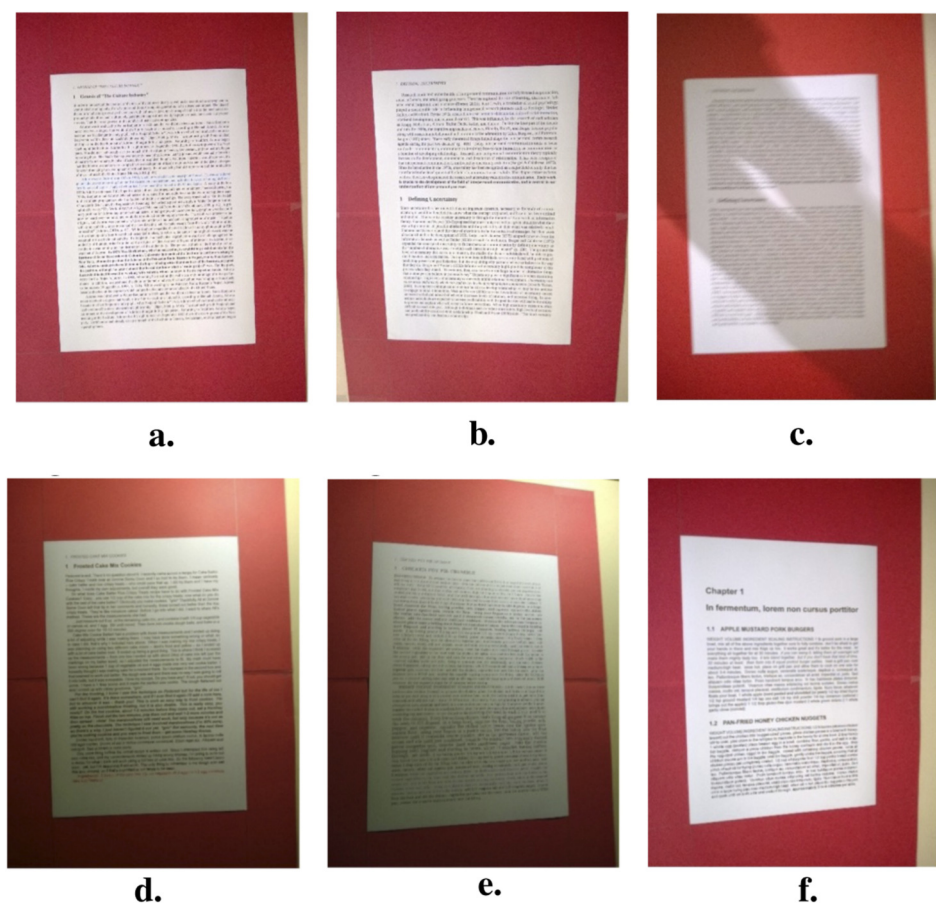


FIGURE 1
(a–f) Sample images from the SmartDoc-QA dataset.

documents across platforms and devices. OCR still remains as a challenging task because of the diverse shapes and scales of texts in the image (Liao et al., 2020). Especially with sequence-like texts, which needs the model to predict a series of labels (Shi et al., 2016).

As the demand for document digitization explodes across industries, the need for faster and more efficient OCR systems becomes ever-more critical. Here, a seemingly mundane aspect—document image quality—emerges as a method that can help in optimizing OCR performance. Image Quality Assessment (IQA) is a specialized field within image processing that focuses on quantifying the quality of an image by measuring how well it retains the information from the original source. This work delves into the realm of document image quality assessment specifically for OCR tasks. By implementing a robust quality assessment mechanism, we can identify and reject images that may hinder character recognition based on predicted OCR accuracy scores without executing OCR system. This will help the research community to retake a photo or scan the document immediately that will save the processing time of document digitization. This selective approach prioritizes the use of high-quality images, ultimately leading to improved accuracy and reliability in OCR output.

In general, IQA is assessing the image in terms of how good it is for visualization. The need for a goal oriented DIQA is to predict how good the document is to recognize the characters without

executing any OCR algorithm. To illustrate this limitation of standard IQA models, consider the visual quality scores obtained from widely used models for a series of document images shown in Figure 1. The Paddle and Keras OCR engines are also executed on the sample images and the accuracy scores are reported in the Table 1 along with the scores of four noreference IQA (NR-IQA) models. The models considered are Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), Natural Image Quality Evaluator (NIQE), Feature maps based Referenceless Image Quality Evaluation Engine (FRIQUEE) and Perceptual Image Quality Assessment (PIQA).

Table 1, shows that there is minimal variance in the quality scores between images as indicated by the NR-IQA models. However, OCR accuracy scores from both Keras OCR and Paddle OCR models exhibit significant variability. In particular, the quality scores for image in the Figures 1a,e are two low it shows that it has poor visual quality but the scores obtained from OCR engine shows that 95% and 73% of the characters in the document are recognized correctly. This stark contrast underscores the necessity for goal oriented IQA methods specifically tailored to document images, aimed at identifying factors hindering character recognition. Hence, the development of dedicated IQA methodologies for document processing is necessary. The processed model assesses the document's image quality in terms

TABLE 1 Comparison on the scores of IQA models and OCR machines for the images displayed in Figure 1.

Sample image	BRISQUE (Mittal et al., 2012a)	NIQE (Mittal et al., 2012b)	FRIQUEE (Ghadiyaram and Bovik, 2017)	PIQA (Venkatanath et al., 2015)	Keras OCR (Morales, 2019)	Paddle OCR (Du et al., 2020)
Figure 1a	0.331	0.046	0.098	0.252	0.293	0.952
Figure 1b	31.38	4.345	10.155	12.257	0.661	0.939
Figure 1c	24.953	4.644	10.535	12.257	0.508	0.982
Figure 1d	23.069	3.647	5.785	6.001	0.312	0.961
Figure 1e	0.108	0.043	0.026	0.060	0.439	0.731
Figure 1f	3.242	2.193	12.195	20.075	0.308	0.953

of the goal that is how good the characters are recognized from the scanned or captured images without executing any OCR engine.

The proposed system tackles document image quality assessment using a goal-oriented approach. When it comes to mobile captured images, there is a high chance of distortions due to camera shake and changes in environment. This can induce blur distortions that can destroy the quality of text in the image (Basar et al., 2023). This approach involves extracting a set of relevant features from the document images that are known to directly impact OCR performance. These features encompass a spectrum of image characteristics, including measures of

- Focus: Focus measures quantify the amount of focus in an image. This can help deal with out of focus pictures.
- Sharpness: The level of detail captured in the image plays a vital role. Low-resolution images may lack the necessary detail for accurate character recognition, particularly for smaller fonts. Sharper images show much more detailed texts since sharper images tend to have more clarity.
- Skew and Rotation: Images with skewed or rotated text present a challenge for OCR systems, requiring additional processing steps that can impact accuracy and efficiency.

By analysing these features, the system aims to predict the expected OCR accuracy score for a given document image. This enables us to filter out unsuitable images before feeding them into an OCR engine, thereby streamlining the OCR workflow and improving overall efficiency.

Ensuring that only high-quality images are used for OCR significantly enhances the accuracy and reliability of digitized content. IQA plays a pivotal role in evaluating and improving the visual content provided to end-users (Zhai and Min, 2020). Accurate OCR ensures that the extracted text closely matches the original document, which is especially critical in domains where data integrity is essential, such as legal documentation, financial records, and medical transcripts. Moreover, high-quality input images help reduce the need for manual intervention, which is often required when OCR systems process poor-quality images. Manual correction of errors or missing information can be time-consuming and resource-intensive. By pre-filtering unsuitable images, the system not only improves OCR accuracy but also minimizes human effort, making document digitization more efficient. Beyond OCR, IQA techniques are also being utilized in

fields such as image enhancement (Li et al., 2023; Alaei et al., 2016) and image restoration (Huang and Xia, 2020), further demonstrating their broad applicability and importance in modern computer vision tasks.

In the context of document IQA (DIQA), reference images are typically not available, making no-reference IQA the most suitable approach. This involves extracting a set of relevant, OCR-focused features—such as focus, sharpness, and geometric distortions—that directly influence text recognition. By analysing these features, the system can predict the quality of document images and enhance OCR performance by filtering out low-quality inputs before processing. In this research, we adopt a DIQA approach, motivated by the limitations of subjective evaluation methods, which depend on human opinion scores and are often time-consuming and inconsistent (Zhou et al., 2024). Instead, our work utilizes OCR accuracy scores generated by modern OCR systems, enabling a more scalable and data-driven assessment of document image quality. The primary objective of this work is to develop a system capable of predicting the OCR accuracy score of a given document image without the need for a reference image.

This paper presents a potential step forward in optimizing OCR workflows. By focusing on document image quality assessment, we can ensure that OCR systems are operating at its highest efficiency, delivering accurate and reliable results across a wide range of document types. The primary objective of this paper is to develop a system capable of predicting the OCR Accuracy score of a given document image, without the need for its reference image. Doing so will help to immediately reject images that are not suitable, that is, that will not give good accuracy result when passed to an OCR system. In essence, our objective is to evaluate the suitability of document images for OCR usage. By implementing this system, we aim to streamline the OCR workflow, improve efficiency, and enhance the accuracy of character recognition for various document types.

To achieve this goal, the following steps are followed

- Feature Engineering: Twelve key features are extracted, including measures of focus, sharpness, and document orientation. These features act as fingerprints, allowing us to understand the image's clarity and suitability for OCR.
- Ground Truth Labelling: The extracted features will be paired with corresponding OCR accuracy scores obtained from two cutting-edge OCR engines: Keras OCR and Paddle OCR. This

labelling process establishes a vital link between image characteristics and OCR performance.

- Machine Learning Model Selection: Leveraging the labelled data, training and evaluation of five Machine Learning models are done. Our objective is to identify the model that demonstrates the most accurate prediction capabilities for OCR accuracy scores on unseen test images.

The remainder of this paper is organized as follows: [Section 2](#) provides a comprehensive review of related work in the areas of DIQA and OCR. [Section 2](#) details the proposed methodology, including the subsections on feature processing, ground truth labelling, and model selection and evaluation. [Section 4](#) presents the performance evaluation of the model on both the public and private datasets. The last section, concludes the research work with a future direction for research.

2 Literature survey

Document quality prediction has posed a longstanding challenge, prompting numerous proposed methods. The study by [Ye and Doermann \(2013\)](#) investigates DIQA, examining existing methodologies. It outlines fundamental steps from creation to processing, highlighting various noise and their impact on optical character recognition. Both Objective and subjective IQA approaches are reviewed, emphasizing their principles and challenges. The study underscores the complexity of document IQA, recognizing the interconnected nature of factors influencing image quality. In conclusion, it stresses the importance of addressing these challenges for the development of more effective IQA methodologies. In the works done by [Alaei et al. \(2015\)](#), [Alaei et al. \(2016\)](#), the authors have proposed using full reference image quality assessment, employing modified gradient magnitude similarity deviation (GMSD) and texture similarity metrics. However, acquiring high-quality reference images proves challenging in real-world scenarios. Consequently, no-reference image quality assessment methods emerged as a better alternative. This can be achieved by either deriving a measure from the images correlating with OCR accuracy scores or mapping OCR accuracy scores with features to predict accuracy scores for test images. We explore literature under both categories in this survey.

[Kumar et al. \(2012\)](#) introduce a sharpness metric called difference of differences. The approach calculates sharpness independently in horizontal and vertical directions using 1-D filtering and edge detection, and then combines the results via a Frobenius norm. [Bui et al. \(2018\)](#) utilize focus and sharpness vectors extracted from image regions to train a two-class Support Vector Machine (SVM) classifier that distinguishes between high- and low-quality images. Document sharpness is inferred based on the margin distances from the SVM decision boundary. Similarly, edge gradient and height-to-width ratio features have been used to estimate normalized word error rates, indicating document quality ([Peng et al., 2011](#)). In another study, character gradients ([Li et al., 2018b](#)) from image text patches are computed and aggregated via pooling to generate a document-level quality score. [Rusinol et al. \(2014\)](#) experiment with 24 focus measures drawn from six different

feature families. These are combined in various ways to determine the best-performing feature set with strong correlation to OCR accuracy. Beyond classical methods, several works leverage deep learning for DIQA. [Li et al. \(2017\)](#) propose an attention-based model combining convolutional neural networks and recurrent neural networks in a glimpse-action framework to focus on character regions. Their model is trained using a reinforcement learning strategy that minimizes a loss function tied to OCR performance. In [Li et al. \(2019\)](#), extend this idea with a three-stage framework comprising text line detection, line-wise quality prediction using deep regression networks, and document-level scoring via weighted pooling based on text region areas.

[Ye and Doermann \(2012\)](#) propose an unsupervised feature learning method that constructs a visual codebook, performs soft-assignment encoding, and uses max-pooling to extract feature vectors. These are fed into a Support Vector Regression (SVR) model for quality prediction. [Xu et al. \(2016\)](#) adopt a Bag-of-Words model enhanced by high-order image statistics (mean, covariance, co-skewness) extracted from clusters formed via K-means. [Peng et al. \(2015\)](#) use Latent Dirichlet Allocation (LDA) to discover document quality topics from raw patches. Later, they extend this with semi-supervised learning ([Peng et al., 2016](#)), incorporating OCR accuracy as supervision to improve the discriminative power of the system. Ridge regression is used for final quality prediction. In deep learning-based approaches, [Kang et al. \(2014\)](#) extract character patches from documents and process them using a CNN with ReLU activations and a specialized max-min pooling layer. The model is trained via stochastic gradient descent to minimize the error between predicted and actual OCR scores. [Lu and Doods \(2019\)](#) employ a transfer learning strategy using AlexNet as the base model. They fine-tune its layers to extract quality-sensitive features and regress OCR scores, optimizing the network with small batch sizes and regularization for improved generalization.

In Common, DIQA methods that use OCR accuracy as a supervision signal tend to perform better, as OCR accuracy serves as a direct measure of a document's readability and content fidelity. For example, it is mentioned in [Rusinol et al. \(2014\)](#) that the method fails in cases where there is a presence of large backgrounds in images. Incorporating OCR accuracy as a label, ensures that the model is trained to prioritize features crucial for accurate OCR processing, leading to more effective quality predictions for real-world document applications. Conversely, DIQA methods that do not use OCR accuracy may overlook important factors affecting document quality, resulting in less accurate predictions that may not align well with practical performance metrics.

An exploration of OCR systems can help with identifying latest OCR models that perform well on document images. OCR implementations can be categorized as machine learning based and deep learning-based methods. Under machine learning based implementations comes the survey by [Jain et al. \(2016\)](#). In this study, a comparison between machine learning models is performed to find out the best performing machine learning model for the MNIST dataset. The work utilizes pixel density of handwritten digits as input. The algorithms produce labels for each digit, which are compared with actual MNIST labels to determine accuracy. Among the algorithms tested, Logistic Regression emerges as the

most accurate for handwritten digits. Comparison with other algorithms highlights Logistic Regression's superiority in classifying hand-drawn digits, while analysis reveals a general trend of decreasing accuracy with increased data size, particularly relevant for multi-class classification tasks.

Existing DIQA techniques span a variety of approaches, including feature extraction (Basar et al., 2023; Zhou et al., 2024), metric-based models (Alaei et al., 2015; 2016), supervised and unsupervised methods (Bui et al., 2018; Li et al., 2017; 2019; Kang et al., 2014; Lu and Dooms, 2019; Jain et al., 2016) based on vision learning. The method suggested in Basar et al. (2023), primarily focus only on defocus blur which limits the method comprehensiveness, as it does not account for other common degradations like motion blur, noise, or uneven lighting, which also impact OCR accuracy. In Zhou et al. (2024), reliance on edge features which also makes it vulnerable to variations in font thickness, background clutter or low-contrast text which can distort edge maps and lead to inaccurate quality estimates. The performance of similarity-based metrics, such as GMSD (Alaei et al., 2015), can be significantly influenced by misalignments and scale inconsistencies between the reference and test images. These variations may introduce artifacts in the similarity computation, resulting in misleading quality scores that do not accurately reflect the perceptual or OCR-relevant degradation in document images. The effectiveness of texture similarity (Alaei et al., 2016) can be compromised by variations in font styles, layout structures, and scanning artifacts, which may not directly correspond to perceptual or OCR-relevant quality degradation. The attention-based recurrent neural network model introduced in (Li et al., 2017) performed a closely tied to its ability to correctly identify and attend to text regions which results in suboptimal quality predictions. The multi-stage architecture suggested in Li et al. (2019) perform high computational time which make the method less suitable for real-time processing and minimal latency. The study by Jain et al. (2016) also observes a general trend of decreasing accuracy with increasing dataset size, which is particularly important for multi-class classification problems. Also, it does not address how machine learning performance varies with DIQA and missing the opportunity to link IQA with OCR output.

To overcome the key limitations of existing DIQA techniques, our proposed method introduces a structured three-module framework encompassing feature extraction, OCR-based labelling and model selection and evaluation for accurate prediction of OCR-relevant image quality. A central component of our framework is the feature extraction process, which is designed to identify the most informative predictors of OCR accuracy. Specifically, we incorporate twelve distinct features, systematically grouped into three categories corresponding to common distortion types encountered during document image acquisition. This comprehensive feature set allows for more precise and reliable evaluation of document image quality. The extracted features are paired with actual OCR accuracy scores derived from two state-of-the-art OCR engines such as Keras OCR and PaddleOCR to confirm quality predictions are closely aligned with real-world text recognition performance. Training the predictive models using a diverse feature set extracted from document images and ground truth labelled with OCR accuracy scores.

Under deep learning implementations, the Paddle OCR, or the PP-OCR system proposed by Du et al. (2020) is explored. Here, The

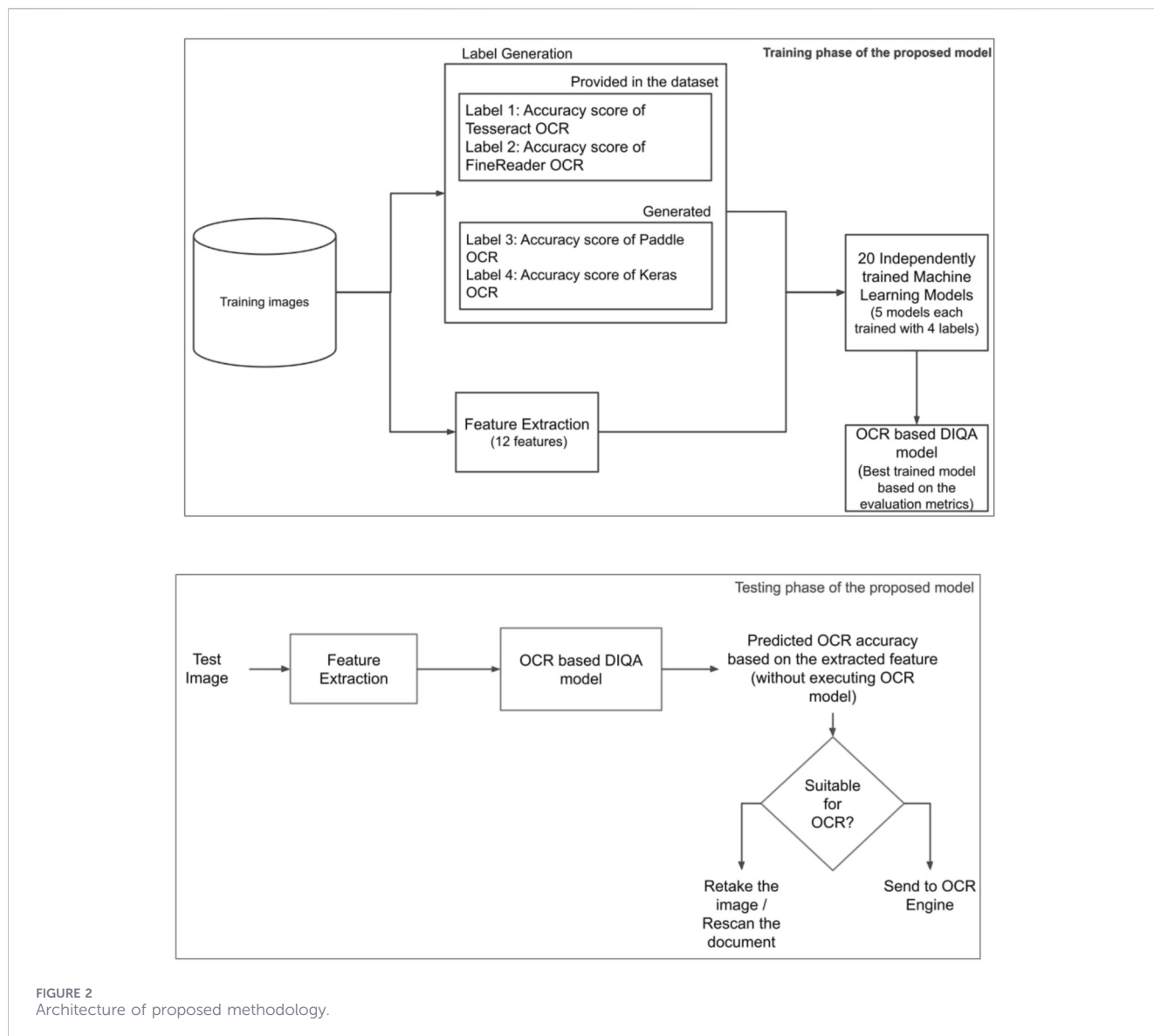
PP-OCR system implements text detection, detection box rectification, and text recognition. Differentiable Binarization (DB) enables efficient text detection, while a text direction classifier ensures correct orientation. Convolutional recurrent neural network (CRNN) is employed for text recognition, optimized with strategies like light backbone and data augmentation. Large-scale datasets support training and evaluation, with ablation experiments are conducted to assess the effectiveness of the strategy. The system is validated for recognition across multiple languages. The Keras OCR (Morales, 2019) is another deep learning implementation that was explored for OCR. This system simplifies text extraction from images via deep learning, employing pre-trained models for text detection and a CRNN architecture for character sequence analysis, leading to accurate text recognition. With Python integration and a user-friendly application programming interface, Keras-OCR serves as a versatile tool bridging text detection and recognition, facilitating automated text extraction tasks effectively. Deep learning implementations excel over machine learning based approaches due to their ability to effectively capture the intricate structures and patterns present in images, facilitated by neural networks. Additionally, assessing model performance with accuracy scores from multiple OCR machines ensures the model's generalizability. Thus, we leverage accuracy scores from both Keras and Paddle OCR machines to enhance the robustness of our predictions.

Existing DIQA techniques have explored various approaches, including metric-based methods, supervised and unsupervised machine learning, and deep learning. While these methods have shown promise, there's room for improvement in DIQA accuracy. Our proposed method trains the models with labels from recent OCR technologies like the Keras OCR and the Paddle OCR. This ensures the DIQA system is aligned with the performance characteristics of modern OCR tools. A critical aspect of feature extraction for DIQA is identifying the optimal feature set that most accurately predicts OCR accuracy scores. Our proposed method tackles this challenge by incorporating twelve diverse features. These features encompass three primary categories of distortions that commonly degrade document images during acquisition.

3 Proposed methods

Figure 2 shows the architecture of proposed model. A machine learning based model was developed to tackle the above mentioned problem. Since OCR accuracy is a numerical value, we train and evaluate five different regression models. These models analyse the labelled and learn to predict the OCR accuracy score for unseen document images. Ultimately, our goal is to identify the most effective model for accurately predicting OCR performance on new documents. Experimented with the following regression model to strength the study. They are Lasso, Ridge, Support Vector, Random Forest and XGBoost regression model.

Lasso regression method combats overfitting, emphasizes model interpretability by penalizing complex models. Its penalty term favors fewer impactful features, driving unimportant ones' coefficients to zero, aiding in feature selection. Lasso excels in datasets with many features, identifying influential factors while maintaining accuracy, empowering data analysts with a potent tool.



Ridge regression (Marquardt and Snee, 1975), enhances linear regression by penalizing complex models, crucial for correlated or large feature sets. It adds a penalty term to the sum of squares of coefficients, reducing reliance on individual attributes, especially useful in multicollinearity cases. By stabilizing coefficients' variance, it fosters more accurate and stable models, improving generalization to new data.

SVR (Awad et al., 2015) utilizes Support Vectors to derive a hyperplane fitting all data points, diverging from conventional regression models. It adjusts error tolerance, identifying the optimal fitting line within specified margins, leveraging principles from SVMs. Random Forest regression (Breiman, 2001) is an ensemble learning model averages predictions from multiple decision trees trained on subsets of data with random feature selection, mitigating overfitting. It aggregates forecasts from all trees to produce the final output, offering versatility in managing extensive datasets and providing insights into feature importance, favored for regression across domains. XGBoost regressor (Chen and Guestrin, 2016), continuous value forecasting model harnesses

ensemble learning by combining strengths from multiple decision trees built sequentially via gradient boosting. Employing regularization to prevent overfitting, XGBoost aggregates predictions from all trees for exceptionally accurate forecasts, handling complex relationships and diverse regression tasks effectively. After training our models, we use a set of metrics (MSE, RMSE, PLCC, SROCC, and R2) to assess their performance. These metrics tell us how well each model predicts OCR accuracy on unseen images.

3.1 Feature extraction

Feature extraction is a critical step in assessing the quality of document images, particularly due to common distortions such as geometric warping, focus degradation, motion blur, and skew. To address these challenges, a total of 12 features are extracted from each image. Of these, 10 features are dedicated to various focus measures, capturing clarity and detail levels. One feature specifically quantifies sharpness, while another evaluates geometric distortion

by measuring skew. Together, these features provide a comprehensive representation of image quality, enabling more accurate prediction of OCR performance.

To comprehensively capture the visual quality of document images, a set of 12 features is extracted: 10 related to focus (Pertuz et al. (2013)), one for sharpness, and one addressing geometric distortion (skew). Below is an overview of each feature as follows: i) Tenengrad Measure uses vertical and horizontal Sobel operators to compute image gradients. The squared gradient values are summed to yield a global sharpness score. High values indicate crisp, well-defined edges and minimal blur, making this useful in applications like microscopy and OCR-related focus evaluation. ii) Tenengrad Variance captures the spatial variance of gradient magnitudes. It reflects the distribution of sharpness across the image, with higher variance implying better focus. iii) Variance of Laplacian quantifies the amount of high-frequency content, such as edges and rapid intensity transitions. It is commonly used to assess the sharpness and focus of an image. iv) Modified Laplacian is obtained by applying a Laplacian filter to the image and computing the mean of the absolute pixel values. It provides localized focus measures and highlights fine details by emphasizing edge regions. v) Modified Laplacian of Gaussian approach first smooths the image using a Gaussian filter and then applies the Laplacian. The combination enhances edge detection and highlights high-frequency details, serving as a strong indicator of image sharpness (Nayef et al., 2015). vi) Brenner's Measure calculates the second-order difference in gray-level intensity between adjacent pixels. It captures the rate of brightness transitions, with higher values indicating sharper edges and clearer image regions. vii) Differences of Median filtered image (Kumar et al., 2012; Bui et al., 2018) is a sharpness metric assesses edge crispness by comparing changes in pixel intensities across successive median-filtered versions of the image. It normalizes sharpness by contrast and aggregates directional changes, providing a robust and noise-resistant measure. viii) Vollath's Measure computes the correlation between an image and its spatially shifted version which quantifies the structural consistency of image content, where higher values indicate greater focus and more coherent text or visual structures. ix) Gradient Energy evaluates the intensity of edges and transitions by summing the squared gradient magnitudes in both directions. It also reflects the overall energy of image gradients, where higher scores imply stronger edge features. x) Squared Gradient derived by squaring the horizontal first derivative, this metric emphasizes sharp transitions and edge prominence in the image. Larger values signify areas with significant detail and clarity. xi) Histogram Range assesses the spread of grayscale values in the image histogram typically correlates with better focus, as it indicates the presence of diverse pixel intensities. It is computed as the difference between the maximum and minimum pixel values. xii) Skew Angle measures the tilt or skew in document images, which often results from improper scanning or camera alignment. Evaluating each rotated version using a scoring function that checks horizontal alignment of text lines. The angle with the highest score is selected as the skew correction angle, and both the angle and corrected image are retained. This helps quantify and correct geometric distortions.

In Figure 3, the heatmap illustrates the pairwise Pearson correlation coefficients among the twelve handcrafted image

quality features extracted from the SmartDoc-QA dataset. Warmer colors indicate higher positive correlation, while cooler tones represent negative or weaker relationships. The visualization helps in identifying redundancy or multicollinearity among features, which is critical for effective feature selection in machine learning-based prediction tasks.

3.2 Labelling the images

While the original dataset includes OCR accuracy labels derived from older OCR systems, we regenerate the labels using two modern OCR engines: PaddleOCR and Keras OCR. Each image is passed through both OCR engines, and the corresponding OCR output is extracted and saved as text files. These outputs are then compared against the ground truth text to compute the OCR accuracy score using the Levenshtein distance, a standard metric for evaluating character-level differences (as shown in the Equation 1).

$$Accuracy = \frac{(A - LevenshteinDistance)}{A} * 100 \quad (1)$$

Where A is the $\max(len(GT), len(OCR))$ GT represents the ground truth, OCR represents the recognized output. Based on the newly generated labels, the evaluation is divided into three parts: i) Evaluation using existing dataset labels to establish a baseline for comparison, ii) Evaluation using OCR scores from PaddleOCR and iii) Evaluation using OCR scores from Keras OCR.

3.3 Model selection and evaluation

Machine learning enables the creation of a goal-oriented DIQA system that directly offers a data-driven solution that learns these complex patterns for accurate and reliable character recognition by modern OCR systems. Since OCR accuracy is a continuous numerical value, the task is formulated as a regression problem. Accordingly, five regression models were implemented and evaluated: Lasso Regression, Ridge Regression (Tibshirani, 1996), SVR (Marquardt and Snee, 1975), Random Forest Regression (Awad et al., 2015), and XGBoost Regression (Breiman, 2001). Each model is trained using extracted features paired with ground-truth OCR accuracy scores obtained from Keras OCR and PaddleOCR. The objective is to learn the relationship between image quality features and OCR performance, enabling accurate prediction of OCR scores for previously unseen document images. Lasso applies L1 regularization to drive coefficients of less important features to zero and effectively performing feature selection. It also identifies the most impactful quality features (e.g., sharpness, skew) that truly influence OCR outcomes, making the model more interpretable. Ridge regression (Tibshirani, 1996) introduces L2 regularization, which excels in high-dimensional spaces and penalizes the sum of squared coefficients to handle multicollinearity. SVR leverages the principles of SVM which captures complex patterns in distortions (blur + skew) leading to more precise predictions in edge cases. The ensemble method (Awad et al., 2015) constructs multiple decision trees using random subsets of the data, enabling the model to capture complex feature interactions. Additionally, it provides insights into feature importance, helping identify which image quality attributes such as focus, sharpness, or skew are most predictive of OCR performance.

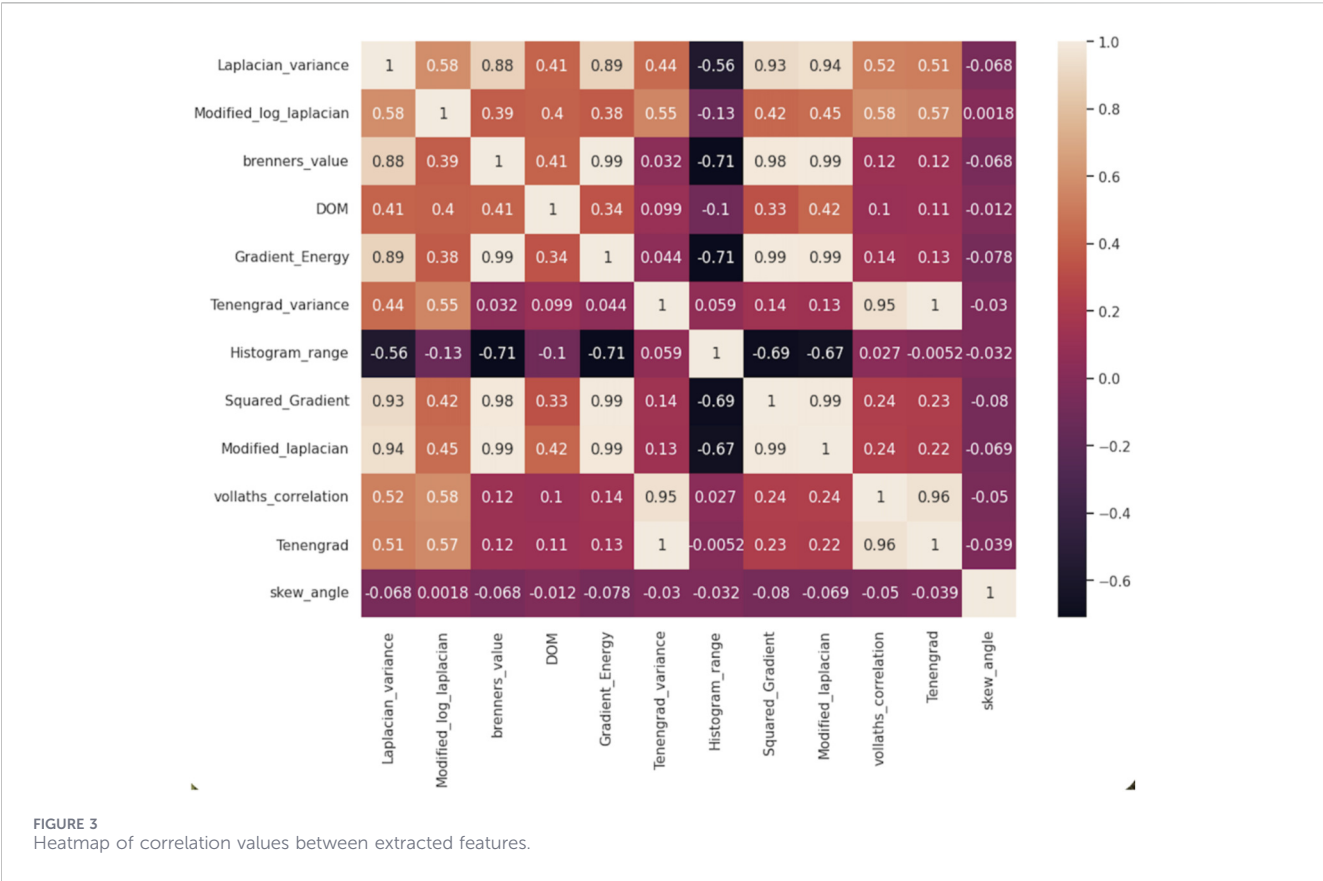


FIGURE 3 Heatmap of correlation values between extracted features.

4 Experimental analysis

This section describes the performance of the model in the public dataset and on a real time dataset. The unique way of labelling the images is also described in this section.

4.1 Dataset description

To effectively predict OCR accuracy from document image quality, it is essential to use a dataset that contains a diverse range of real-world distortions typically encountered when capturing documents using smartphones. For this study, we utilize the SmartDoc-QA dataset (Chen and Guestrin, 2016), which contains 2130 images. These images are generated by capturing 30 documents from 71 different angles, introducing a variety of distortions such as blur (out-of-focus), lighting variations, geometric distortions, shadows, and more. The dataset includes the ground truth text for each document as well as OCR outputs and accuracy reports generated using two popular OCR systems: Tesseract and ABBYY FineReader. For our experiments, we extract twelve image quality features from each document image. These features represent various aspects of image clarity and alignment that affect OCR performance. A total of 1598 images are used for training, while the remaining 532 images are reserved for testing and evaluation. All machine learning models are initially trained using default hyperparameters. The evaluation compares the impact of different labelling strategies to determine which approach yields the most accurate predictions.

4.2 Performance analysis of the model

The SmartDoc-QA dataset includes OCR accuracy reports generated by two traditional OCR engine using Tesseract OCR and ABBYY FineReader. These reports indicate the recognition accuracy of the extracted text for each document image. To maintain consistency and enable direct comparison, the OCR accuracy scores are standardized to a range between 0 and 1 and assigned as labels to their corresponding images. The entire dataset is then used to train five different machine learning models such as Lasso Regression, Ridge Regression, SVR, Random Forest Regression, and XGBoost Regression. Each model is trained to predict OCR accuracy scores on a separate test set composed of unseen document images. Model performance is evaluated by comparing the predicted OCR scores with the actual standardized scores in the test set using several key metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Pearson Correlation Coefficient (PCC), Spearman Rank Order Correlation Coefficient (SROCC) and R-squared (R^2). These metrics collectively quantify both the prediction error and the strength of correlation between the actual and predicted values. It's important to note that in this study, "actual values" refer to the OCR accuracy scores directly obtained from the OCR engines.

4.2.1 Accuracy score of tesseract OCR model as label

Table 2 summarizes the performance of five regression models such as Lasso, Ridge, SVR, Random Forest, and XGBoost trained

TABLE 2 Evaluation results for labelling with tesseract OCR.

Model name/Paper title	PCC	SROCC	MSE	RMSE	R2 score
Towards document image quality assessment: A text line based framework and a synthetic text line image dataset Li et al. (2019)	0.7433	0.7572	NA	NA	NA
Attention based RNN model for document image quality assessment Li et al. (2017)	0.814	0.865	NA	NA	NA
SEDIQA Courtney (2021) , Alaei et al. (2023)	0.8463	0.8582	NA	NA	NA
TextNet Li et al. (2018a) , Alaei et al. (2023)	0.6840	0.6730	NA	NA	NA
Proposed features with ridge regression	0.6958	0.7271	0.0406	0.2015	0.4799
Proposed features with lasso regression	0.6998	0.7308	0.0404	0.2009	0.4832
Proposed features with SVR	0.8383	0.8039	0.0236	0.1537	0.6975
Proposed features with random forest	0.8416	0.8207	0.0229	0.1512	0.7072
Proposed features with XGBoost	0.8533	0.8251	0.0213	0.1460	0.7270

Best scores are highlighted in bold.

TABLE 3 Evaluation results for labelling with Finereader OCR.

Model name/Paper title	PCC	SROCC	MSE	RMSE	R2 score
Towards document image quality assessment: A text line based framework and a synthetic text line image dataset Li et al. (2019)	0.7433	0.7572	NA	NA	NA
Attention based RNN model for document image quality assessment Li et al. (2017)	0.814	0.865	NA	NA	NA
SEDIQA Courtney (2021) , Alaei et al. (2023)	0.8463	0.8582	NA	NA	NA
TextNet Li et al. (2018a) , Alaei et al. (2023)	0.6840	0.6730	NA	NA	NA
Proposed features with ridge regression	0.7719	0.8021	0.0646	0.2542	0.5958
Proposed features with lasso regression	0.7814	0.8082	0.0623	0.2496	0.6101
Proposed features with SVR	0.8506	0.8362	0.0449	0.2117	0.7195
Proposed features with random forest	0.8540	0.8485	0.0447	0.2113	0.7207
Proposed features with XGBoost	0.8528	0.8440	0.0455	0.2132	0.7156

Best scores are highlighted in bold.

using Tesseract OCR accuracy scores as ground truth labels. The models are evaluated based on various metrics such as MSE, RMSE, PCC, SROCC and R^2 and the results are compared with the existing methods and particularly XGBoost and Random Forest achieved the highest PCC values of 0.8533 and 0.8416, respectively. These values outperform the state of the art techniques by highlighting the strong predictive power of these ensemble-based models. However, the SROCC scores for both models were slightly lower than the values reported in [Li et al. \(2017\)](#), suggesting that the models accurately capture the general trends of OCR accuracy, their ability to predict the precise ranking of image quality still leaves room for refinement.

4.2.2 Accuracy score of FineReader OCR model as label

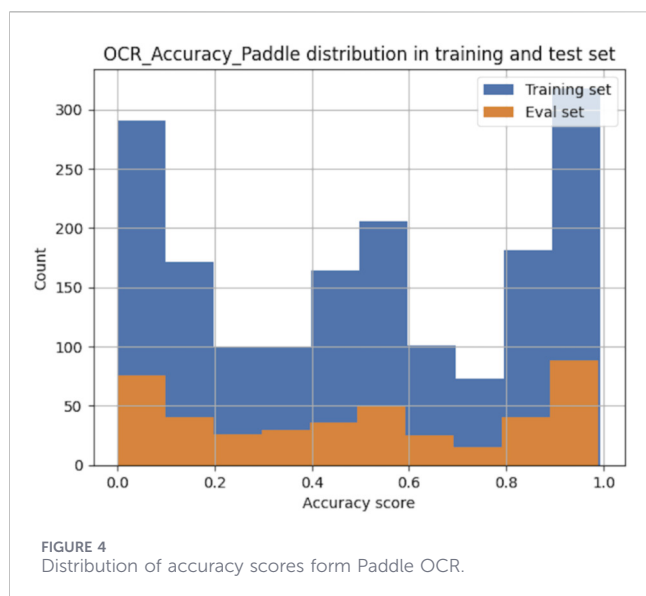
When the FineReader OCR accuracy scores are used as labels, the evaluation results shown in [Table 3](#) indicate marginally lower performance across all models. In this case, the Random Forest regressor emerged as the top performer, with a PCC of 0.8540, indicating a strong positive correlation between the predicted and

actual accuracy scores. Similar to the Tesseract labels, the Spearman Rank Correlation Coefficient (SROCC) score was lower (0.8485) compared to the value reported in [Li et al. \(2017\)](#). This suggests that while our model excels at predicting the overall trend of OCR accuracy for Finereader, it still has room for improvement compared to existing methods.

4.2.3 Accuracy score of paddle OCR model as label

Paddle OCR ([Du et al., 2020](#)) developed in 2020, is a fast, multilingual, lightweight OCR system excelling at text recognition. It operates in three steps. The first step involves text detection using a small segmentation network. In the second step, text box rectification is performed with a compact classifier. Finally, text recognition is performed using a CRNN architecture. The authors built large datasets for training and offer an easy-to-use pre-trained Python library for developers. For this research, we leverage the pre-trained English recognition model.

Once we extract key features from the document images, we label this data by incorporating the OCR accuracy scores obtained



from the Paddle OCR model. The scores are then scaled to the range 0–1. These scores act as labels, essentially linking the image characteristics to their expected OCR performance. This combined dataset is then fed to various regression models for training. The histogram in Figure 4 shows the distribution of OCR accuracy scores obtained from the Paddle OCR machine for the SmartDoc dataset. The x-axis represents the accuracy, ranging from 0.0 to 1.0, with 0.0 indicating full error and 1.0 indicating perfect accuracy. The y-axis represents the number of scores that fall within each accuracy range. We see that majority of the values are either near 0 or near 1.

Table 4 shows the results from the five models when the Paddle OCR accuracy scores are used as label for the dataset. The first two rows in the table are existing methods, while the rest of the rows shows the metric values for the regression models. Random Forest and XGBoost emerge as the top performers. Both achieve similar PCC values around 0.77–0.79 and SROCC values around 0.77–0.78, demonstrating a good ability to predict the general trend in OCR accuracy. Additionally, their MSE and RMSE values are lower compared to other models, suggesting a smaller average error in their predictions. Ridge Regression, on the other hand, shows the lowest performance among the evaluated models. Its PCC score of 0.6177 and SROCC score of 0.6373 is considerably lower, suggesting

limitations in both capturing the general trend and the relative ranking of image quality for OCR accuracy prediction.

4.2.4 Accuracy score of keras OCR model as label

Keras OCR is a deep learning-based framework that streamlines text extraction from images through a structured multi-stage process. The pipeline begins with text detection, where models such as Character Region Awareness for Text Detection (CRAFT) identify and generate bounding boxes around text regions. Following detection, each region is passed to a CRNN, which combines convolutional layers for feature extraction with recurrent layers for sequential character analysis. The CRNN predicts the character sequences present within the detected text regions. In the final stage, post-processing is applied to refine the raw outputs, enhancing the accuracy and readability of the recognized text. Keras-OCR is particularly known for its seamless Python integration, user-friendly API, and flexibility for custom training, making it a powerful and adaptable tool for automated document text recognition. This cohesive pipeline effectively bridges the gap between text detection and recognition, ensuring a robust OCR performance across varied image conditions.

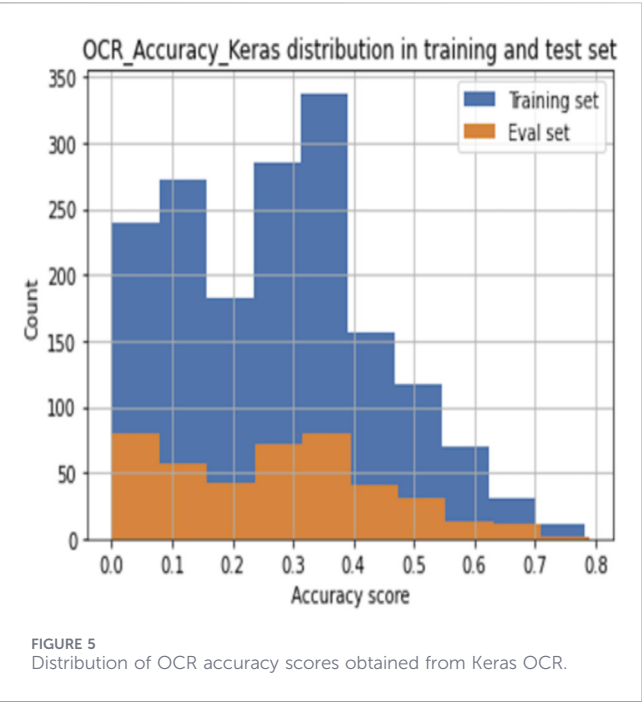
Figure 5 contains the histogram showing the distribution of the accuracy scores of the document images, after character recognition from the Keras OCR machine. The data leans towards the left side of the graph, indicating that a significant portion of the images received accuracy scores concentrated around the value 0.35.

When these accuracy scores are used as label for the image, the results from the five models are summarized into the Table 5. The values in the table suggest a very strong positive correlation between the predicted and the original labels, meaning that the models are predicting the values accurately. Between the five proposed models, XGBoost emerges as the best performer with a PCC of 0.9139, SROCC of 0.9193 and a RMSE of 0.0684. This signifies that, on average, the predictions made by the model deviated from the actual values by only 0.0684 units. This explains the high correlation between the original and the predicted values. Additionally, we observe that the XGBoost regressor shows better results when compared to the existing models taken for comparison, in terms of both PCC and SROCC. This suggests that this method of labelling is quite effective in comparison to cases 1 and 2. This also suggests that the combination of twelve features extracted are strong features that effectively quantify the factors that affect text recognition in document images.

TABLE 4 Evaluation results for labelling with Paddle OCR.

Model name	PCC	SROCC	MSE	RMSE	R2 score
Proposed features with ridge regression	0.6177	0.6373	0.0650	0.2549	0.3778
Proposed features with lasso regression	0.6286	0.6455	0.0639	0.2526	0.3888
Proposed features with SVR	0.7376	0.7419	0.0498	0.2230	0.5238
Proposed features with random forest	0.7773	0.7750	0.0421	0.2052	0.5966
Proposed features with XGBoost	0.7853	0.7811	0.0422	0.2054	0.5960

Best scores are highlighted in bold.



In order to comprehensively evaluate the efficacy of our model in comparison to established general NRIQA (No-Reference Image Quality Assessment) models, we have generated a detailed graph, as depicted in Figure 7. This graph illustrates the relationship between the predicted values obtained from the XGBoost regressor and the corresponding values generated by four existing NRIQA models. It is worth noting that prior to plotting, all output values have been uniformly scaled to the range of 0–1 for consistency. Additionally, in the case of the PIQE model—wherein a lower score indicates higher image quality—we have taken the measure of subtracting each value from 1. This adjustment ensures that lower values across all models uniformly signify diminished image quality.

Upon scrutiny of the graph, a discernible pattern emerges: the assessment of image quality varies significantly across different methodologies employed by the NRIQA models. This observation underscores the nuanced nature of quality assessment, wherein diverse approaches yield disparate evaluations for the same image. Consequently, it becomes evident that the quality labels provided by the NRIQA models may not exhibit strong correlation with the OCR accuracies, as opposed to the predictions generated by our XGBoost regressor. Therefore, the

output derived from the XGBoost regressor emerges as the optimal choice when compared with the outputs from the existing NRIQA models, signifying its superior performance in the realm of image quality assessment. As a summary, the XGBoost regressor demonstrates the highest performance among the models, particularly when considering Case 3 of labelling, which utilizes OCR accuracy from Keras OCR.

4.3 Performance of the proposed model in private dataset

To further validate our model’s generalization capability, we extended the evaluation to a private dataset containing unseen document images. To assess the real-world applicability of our model, we curated a private dataset consisting of 100 document images. These images represent 20 unique documents including books, handwritten notes, and shopping receipts. Each document images are captured under five distinct conditions to simulate common distortions such as geometric misalignment and blur, typically encountered during smartphone-based scanning. All images were taken in a controlled indoor setting using artificial white lighting, without influence from natural light. The device used to capture the document images is the Redmi 10 mobile phone. It has the following camera specifications: 50 MP main camera, 8 MP ultrawide camera, 2 MP macro camera and 2 ML depth sensor.

These images were taken under the default settings of the camera. Subsequently, the captured images were processed through two OCR models, and the resulting output was compared with the ground truth to determine the accuracy of character recognition. Following this, the 12 features extracted from the images were labelled with the corresponding OCR accuracy scores. The dataset, thus prepared, was then fed into the five models for evaluation. For training and testing purposes, 80% of the private dataset was allocated for training, while the remaining 20% was reserved for testing. Figure 6 shows a sample of 6 images from the private dataset.

Upon evaluation on the test set, the model that exhibited the highest performance was XGBoost, attaining a PCC of 0.7613. This result was observed specifically when employing the accuracy scores from the Keras OCR as the label. Interestingly, when utilizing the Paddle label, the model’s performance slightly declined with a PCC value of 0.3911, a trend similarly noted in cases where the SmartDoc dataset was employed. This observation aligns with the findings obtained from the SmartDoc dataset, thereby validating the consistency of results across different labelling methodologies.

TABLE 5 Evaluation results of the five regression models using OCR accuracy scores from Keras OCR as labels.

Model name	PCC	SROCC	MSE	RMSE	R2 score
Proposed features with ridge regression	0.7272	0.7574	0.0131	0.1143	0.5285
Proposed features with lasso regression	0.7340	0.7649	0.0128	0.1132	0.5372
Proposed features with SVR	0.8641	0.8763	0.0071	0.0845	0.7424
Proposed features with random forest	0.9096	0.9148	0.0049	0.0698	0.8239
Proposed features with XGBoost	0.9139	0.9193	0.0047	0.0684	0.8311

Best scores are highlighted in bold.

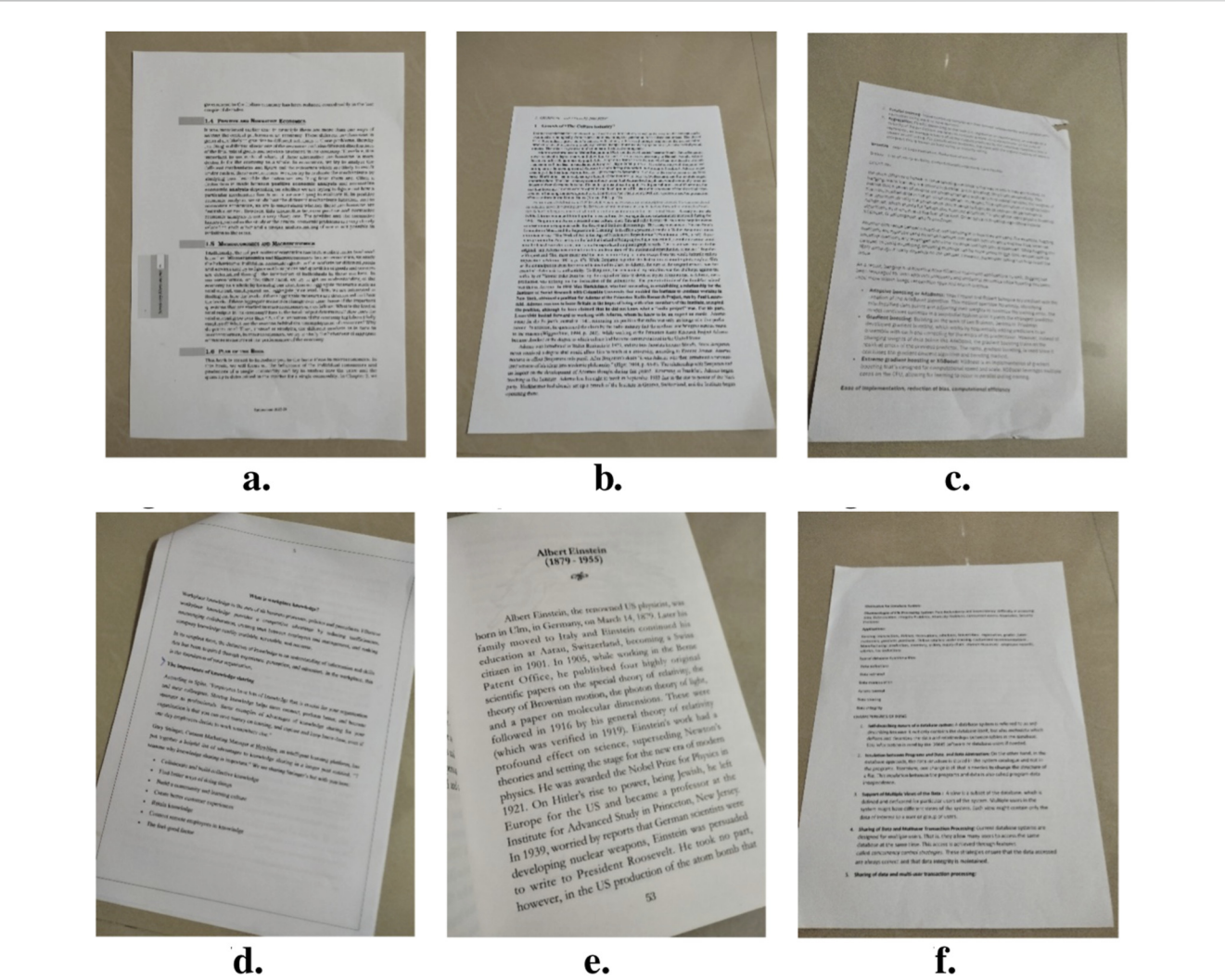


FIGURE 6 (a-f) Sample images from private dataset.

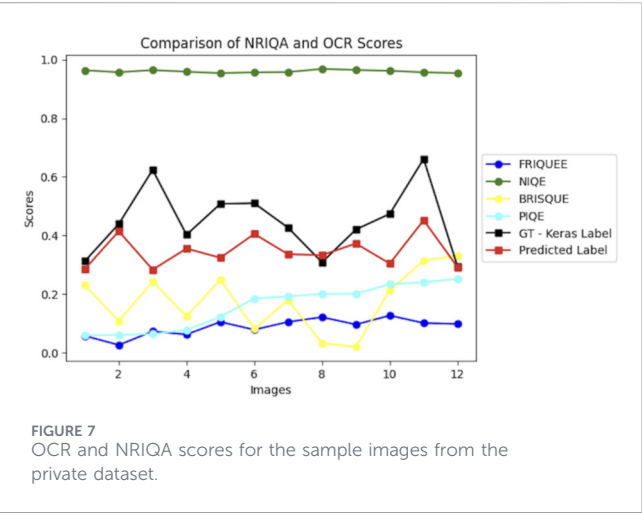


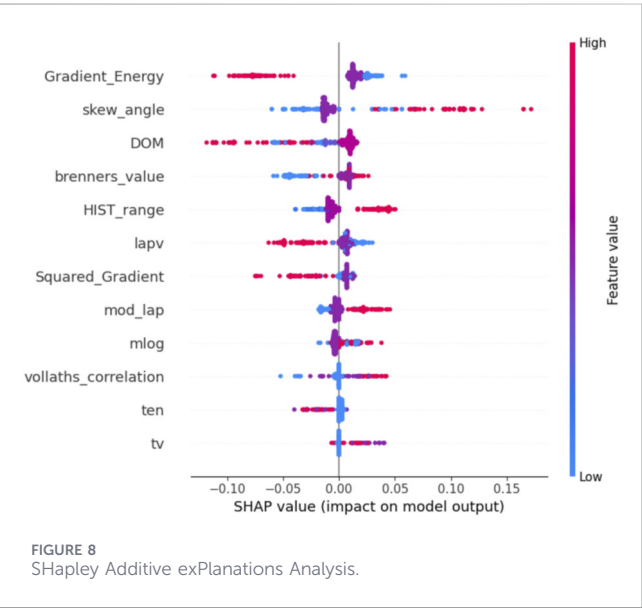
FIGURE 7 OCR and NRIQA scores for the sample images from the private dataset.

The average of predicted values stands at 0.3884, while the average of values from the OCR machine is slightly higher at 0.4435. The difference between the average values is 0.06615. Notably, there is minimal disparity between these two averages, indicating that the model predicts with remarkably low error. Figure 7 shows the scatter plot of actual and predicted values, for a sample of 12 images from the dataset. Notably, the gap between the actual value, indicated by blue dot, and the predicted value, indicated by a red square, is quite less for most of the images. This observation underscores the accuracy and precision of the model in forecasting OCR accuracy scores based on document image features.

In comparison to the model's performance on the SmartDoc dataset, the PCC observed in the private dataset is slightly lower. This discrepancy may stem from the smaller training size of only 80 images utilized in the private dataset. Increasing the dataset size by incorporating additional images could potentially enhance the performance of the regressor model. By providing the model with a

TABLE 6 Analysis on the contribution of features.

Features	MSE	RMSE	R2 Score	PCC	SROCC
Excluding gradient based features	0.0052	0.0720	0.8014	0.8949	0.9003
Excluding tenengrad family features	0.0074	0.0854	0.7245	0.8475	0.8462
Excluding Laplacian variance	0.0045	0.0670	0.8264	0.9068	0.9114
Excluding histogram range	0.0049	0.0697	0.8131	0.8997	0.9027
Including all the features	0.0046	0.0678	0.8228	0.9048	0.9101



more extensive and diverse dataset, with proper distribution of labels, it can better capture the underlying patterns and nuances present in document images, consequently improving its predictive capabilities.

Overall, the XGBoost regressor stands out as the most effective model across both datasets. It achieved its highest performance with a PCC of 0.9239 and an RMSE of 0.0684 when trained on the SmartDoc dataset labeled using Keras OCR accuracy scores.

4.4 Ablation study

An ablation analysis is performed to show how well the extracted features are observed attributes of the document images in terms of OCR performance. Based on the heatmap shown in the Figure 3, the features are grouped into four different set. They are gradient based, tenengrad family, Laplacian variance and histogram range. Features in the gradient based are brenners values, gradient energy, squared gradient and modified Laplacian. Tenengrad, tenengrad variance and vollaths correlation features are under tenengrad family. 100-fold cross validation performed on each feature group. Table 6 shows the performance of complete model with all the features and the models with reduced features. The statistical significant test for RMSE scores between the model with all feature and model trained by excluding gradient based feature is with the p value of 0, shows the features are statistically significant. Figure 8 shows the SHapley

Additive exPlanations analysis of the features. From the figure its visually proven that all the features are contributing to the model, where some of the features are strongly correlated with prediction. The confidence intervals are calculated for the metrics to show that the model is constantly providing the better performance. The confidence interval for the RMSE score is that 0.0678 ± 0.169 , for PCC 0.9048 ± 0.026 and for SROCC 0.9101 ± 0.026 .

5 Conclusion

Predicting document quality has long been a challenging task, demanding innovative solutions to improve OCR accuracy and efficiency. This project addressed the issue by developing machine learning models capable of accurately forecasting OCR accuracy based on document image quality. By leveraging modern OCR systems—Paddle OCR and Keras OCR—for labelling, and extracting twelve key features related to focus, motion blur, and skew, we constructed robust predictive models. Among the five models evaluated, the XGBoost regressor consistently outperformed others, demonstrating strong predictive accuracy across different datasets. This confirms the feasibility and effectiveness of using machine learning for document quality assessment. While the results are promising, they also reveal areas for further enhancement. Future work could explore additional image features that influence OCR performance, and integrate advanced deep learning techniques to boost model generalizability and accuracy. Embracing such innovations could significantly advance the field of document image quality prediction, paving the way for more intelligent and reliable OCR systems.

Data availability statement

The datasets used for this study are publicly available. This data can be found here: <http://smartdoc.univ-lr.fr/smartdoc-qa/>.

Author contributions

RK: Formal Analysis, Writing – original draft, Data curation, Visualization, Validation. JJ: Investigation, Resources, Writing – review and editing, Project administration. SK: Methodology, Investigation, Conceptualization, Validation, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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